Understanding observability metrics: Types, golden signals, and best practices

1. What is Observability?

Observability means **monitoring and understanding what is happening inside a system using data**.

It helps to:

- Detect problems
- Monitor system health
- Troubleshoot issues

3 Pillars of Observability

1. Metrics

Metrics are **numerical measurements that show system performance over time**.

Examples: CPU usage, Memory usage, Request rate, Error rate

Metrics help teams:

- Monitor system health
- Detect performance problems
- Create alerts and dashboards.

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2. Logs

Logs are **records of events generated by applications or systems**.

They show:

- What happened
- Error messages

- System activities

Logs help engineers **debug problems and understand system behavior**.

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3. Traces

Traces track **the path of a request as it moves through different services** in a system.

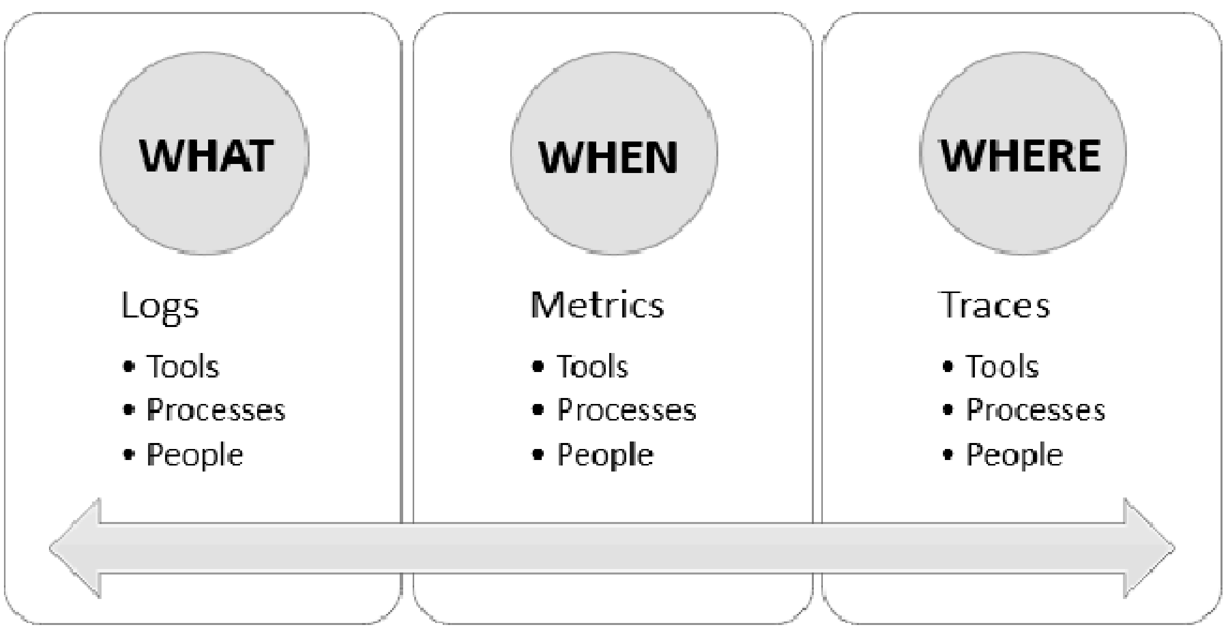
They help:

- Understand service dependencies
- Find slow services
- Identify bottlenecks in microservices architectures

Metrics → show system trends and performance

Logs → provide detailed event information

Traces → show how requests move across services



4 Golden Signals for SRE Teams

The **4 Golden Signals** are key metrics used by **Site Reliability Engineering (SRE)** teams to monitor system health and performance. They help quickly detect problems in applications and services.

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1. Latency

Latency is the **time taken to process a request**.

Example:

- API response time
- Database query time

If latency is high, the system becomes **slow for users**.

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2. Traffic

Traffic shows **how much demand the system is receiving**.

Examples:

- Number of requests per second
- Number of users accessing the service

High traffic means the system is handling **more load**.

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3. Errors

Errors represent the **number of failed requests**.

Examples:

- HTTP 500 errors
- Failed API calls

A high error rate means the **service is not working correctly**.

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4. Saturation

Saturation shows **how much system resources are being used**.

Examples:

- CPU usage
- Memory usage
- Disk usage

High saturation means the system is **close to its limit**.